



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Main)
FULL SYLLABUS
18-08-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of **90** questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer and 0 in all other cases.
5. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited**.
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
7. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
8. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
9. **Do not fold or make any stray marks on the Answer Sheet.**
10. Take $g = 10 \text{ m/s}^2$ unless otherwise stated.

DO NOT BREAK THESE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Name of the Candidate (in Capitals)_____

Form Number : in figures_____

: in words_____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in JEE(Main) 2021

Corporate Office : **ALLEN** CAREER INSTITUTE, "SANKALP", CP-6, Indra Vihar, Kota (Rajasthan) INDIA-324005

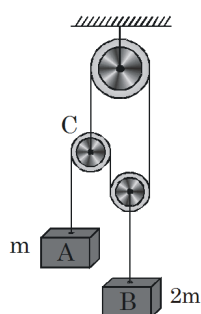
☎ +91-744-2757575 ✉ info@allen.ac.in 🌐 www.allen.ac.in

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:
Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : -1 If wrong option is selected.

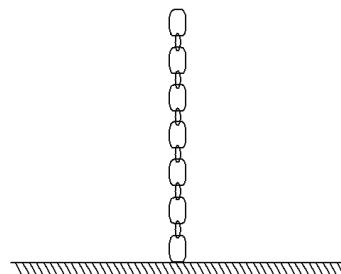
1. In the arrangement shown the pulleys and the strings are ideal. The respective accelerations of the blocks A and B are



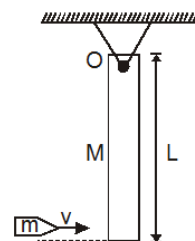
- (A) $g, g/2$ (B) $g/2, g$
 (C) $3g/2, 3g/4$ (D) g, g
2. The potential energy of a diatomic molecule is given by $U = \frac{A}{r^{12}} - \frac{B}{r^6}$. A and B are positive constants. The distance r between them at equilibrium is :-

- (A) $\left(\frac{A}{B}\right)^{1/6}$ (B) $\left(\frac{2A}{B}\right)^{1/6}$
 (C) $\left(\frac{A}{2B}\right)^{1/6}$ (D) None of these

3. A uniform chain of mass m and length ℓ hangs on a thread and touches the surface of a table by its lower end. Find the force exerted by the table on the chain when half of its length has fallen on the table. The fallen part does not form heap.

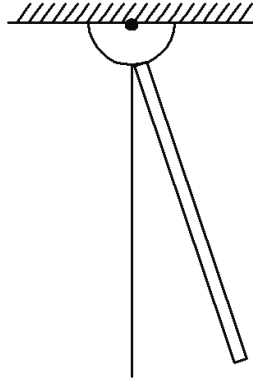


- (A) $\frac{3mg}{4}$ (B) $\frac{mg}{2}$
 (C) $\frac{3mg}{2}$ (D) mg
4. A rod of mass M and length L is suspended by a frictionless hinge at the point O as shown in figure. A bullet of mass m moving with velocity v in a horizontal direction strikes the end of the rod and gets embedded in it. The angular velocity of the rod just after the collision is -

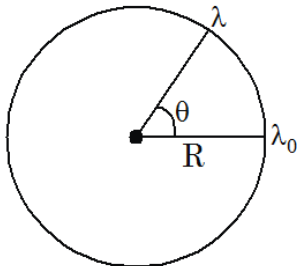


- (A) $\frac{mv}{(3m + M)L}$ (B) $\frac{2mv}{ML}$
 (C) $\frac{2mv}{L}$ (D) $\frac{3mv}{(3m + M)L}$

5. A metre stick swinging in vertical plane about a fixed horizontal axis passing through its one end undergoes small oscillation of frequency f_0 . If the bottom half of the stick were cut off, then its new frequency of small oscillation would become :



- (A) f_0 (B) $\sqrt{2} f_0$
(C) $2f_0$ (D) $2\sqrt{2} f_0$
6. A thin non-conducting ring of radius R has a linear charge density $\lambda = \lambda_0 \cos \theta$, where λ_0 is the value of λ at $\theta = 0$. Find the net electric dipole moment for this charge distribution.

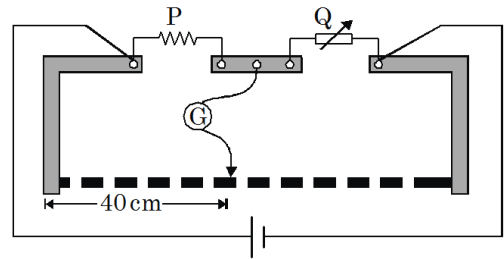


- (A) $\frac{\pi R^2 \lambda_0}{4}$ (B) $\frac{\pi R^2 \lambda_0}{2}$
(C) $\pi R^2 \lambda_0$ (D) $2\pi R^2 \lambda_0$

7. A body is projected away from the earth's surface with a speed $3v_e$ where v_e is the escape velocity. The speed of the body at infinity will be :

- (A) v_e (B) $\sqrt{2} v_e$ (C) $2v_e$ (D) $2\sqrt{2} v_e$

8. In a metre bridge, the gaps are closed by two resistances P and Q and the balance point is obtained at 40 cm. When Q is shunted by a resistance of 10Ω , the balance point shifts to 50 cm. The values of P and Q are :-



- (A) $\frac{10}{3}\Omega, 5\Omega$ (B) $20\Omega, 30\Omega$
(C) $10\Omega, 15\Omega$ (D) $5\Omega, \frac{15}{2}\Omega$

9. A parallel plate capacitor having capacitance C_0 is connected to a battery of emf E . It is then disconnected from the battery and a dielectric slab of dielectric constant k completely filling the air gap of the capacitor is inserted in it. If ΔU indicates the change in energy, then

- (A) $\Delta U = 0$
(B) $\Delta U = \frac{1}{2} C_0 E^2 (k - 1)$
(C) $\Delta U = \frac{1}{2} C_0 E^2 \left(1 - \frac{1}{2k}\right)$
(D) $\Delta U = \frac{1}{2} C_0 E^2 \left(\frac{1}{k} - 1\right)$

10. Given a ring of mass M , radius R resting on a rough horizontal plane, with coefficient of friction ' μ '. A cylindrical, vertical magnetic field passing through ring is switched on at $t = 0$ which varies with time as $B = Kt^2$. Ring has charge Q distributed uniformly over it. Find the time after which ring starts rotating.

- (A) $\frac{\mu Mg}{QKR}$ (B) $\frac{2\mu Mg}{QKR}$
(C) $\frac{\mu Mg}{2QKR}$ (D) $\frac{\mu Mg}{4QKR}$

11. Two moving coil galvanometers M_1 and M_2 having same spring constants have the following specifications:

for galvanometer M_1 , $R_1 = 10\Omega$, $N_1 = 30$,

$A_1 = 3.6 \times 10^{-3} \text{ m}^2$, $B_1 = 0.25 \text{ T}$

for galvanometer M_2 , $R_2 = 14\Omega$, $N_2 = 42$,

$A_2 = 1.8 \times 10^{-3} \text{ m}^2$, $B_2 = 0.5 \text{ T}$

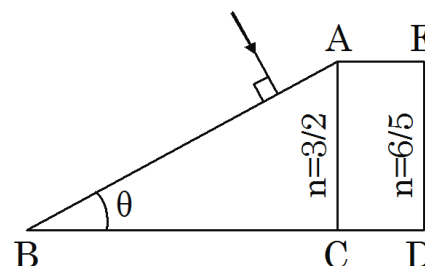
Find the ratios of current sensitivity between galvanometer M_1 & galvanometer M_2

- (A) $5/7$ (B) $3/4$
(C) 1 (D) $7/5$

12. The wavelength of the K_α line for an element of atomic number 57 is α . What is the wavelength of the K_α line for the element of atomic number 29?

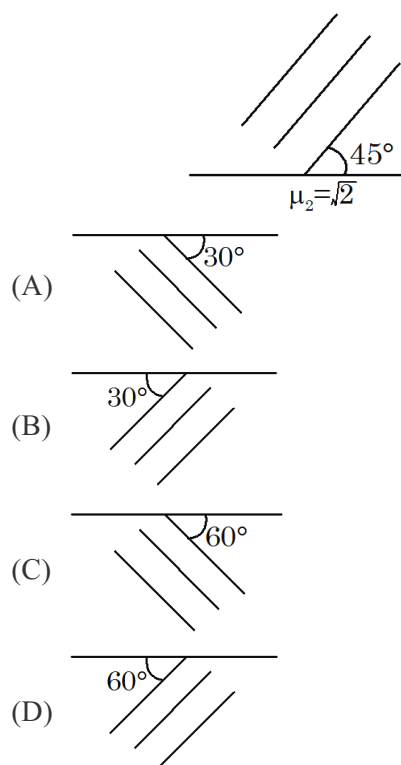
- (A) α (B) 2α
(C) 4α (D) 8α

13. In the figure ABC is the cross section of a right angled prism and ACDE is the cross section of a glass slab. The value of θ so that light incident normally on the face AB does not cross the face AC is (Given $\sin^{-1}(3/5) = 37^\circ$)



- (A) $\theta < 30^\circ$ (B) $\theta < 37^\circ$
(C) $\theta < 60^\circ$ (D) $\theta < 53^\circ$

14. Wavefronts incident on an interface between the media are shown in the figure. The refracted wavefront will be as shown in



15. A fully filled hemispherical tank of radius R has an orifice of small area a at its bottom. Time required to completely empty the tank will be (Assuming that the top surface area of the liquid is always much greater than the orifice area)

(A) $\frac{5\pi}{\sqrt{2}} \frac{R^{5/2}}{ag}$ (B) $\frac{9}{\sqrt{2}} \frac{\pi R^{5/2}}{ag}$
 (C) $\frac{7\sqrt{2}}{15a\sqrt{g}} \pi R^{5/2}$ (D) Can't be calculated

16. The power radiated by a black body is P and it radiates maximum energy around the wavelength λ_0 . If the temperature of the black body is now changed so that it radiates maximum energy around wavelength $3/4\lambda_0$, the power radiated by it will increase by a factor of

(A) $4/3$ (B) $16/9$
 (C) $64/27$ (D) $256/81$

17. Consider the following data:

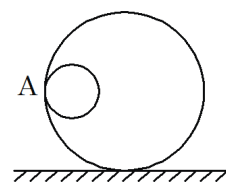
10 main scale division = 1 cm, 10 vernier division = 9 main scale divisions. Zero of vernier scale is to the right of the zero marking of the main scale with 6 vernier divisions coinciding with main scale divisions. The measured reading for length measurement is 4.3 cm on main scale with 2 vernier divisions coinciding with main scale graduations. Estimate the actual length.

(A) 426 cm (B) 4.26 cm
 (C) 4.38 cm (D) 4.5 cm

18. A spring-mass system has undamped natural angular frequency $\omega_0 = 100 \text{ rad s}^{-1}$. The solution $x(t)$ at critical damping is given by $x(t) = x_0(1 + \omega_0 t) e^{-\omega_0 t}$, where x_0 is a constant. The system experiences the maximum damping force at time

(A) 0.01 s (B) 0.1 s
 (C) 0.01 π s (D) 0.1 π s

19. A ring of radius $3a$ is fixed rigidly on a table. A small ring whose mass is m and radius a , rolls without slipping inside it as shown in the figure. The small ring is released from position A. When it reaches at the lowest point, the speed of the centre of the ring at that time would be -



(A) $\sqrt{2ga}$ (B) $\sqrt{3ga}$
 (C) $\sqrt{6ga}$ (D) $\sqrt{4ga}$

20. At any point $(x, 0, 0)$ the electric potential V is $\left(\frac{1000}{x} + \frac{1500}{x^2} + \frac{500}{x^3} \right)$ volt, then electric field at $x = 1 \text{ m}$ -

(A) $5500 (\hat{j} + \hat{k}) \text{ V/m}$
 (B) $5500 \hat{i} \text{ V/m}$
 (C) $\frac{5500}{\sqrt{2}} (\hat{j} + \hat{k}) \text{ V/m}$
 (D) $\frac{5500}{\sqrt{2}} (\hat{i} + \hat{k}) \text{ V/m}$

SECTION-II : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

1. The tangential acceleration of a particle in a circular motion of radius 2m is $a_t = \alpha t$ m/s² (where α is a constant). Initially the particle is at rest. Net acceleration of the particle makes 45° with the radial acceleration after 2 sec. The value of constant α (in m/s³) is:
2. A bomb is projected upwards. At topmost point it explodes in three identical fragments. First fragment comes to ground in 10 sec. and others in 20 sec each. Then the height (in m) reached by the original bomb is :
3. A heavy nucleus having mass number 200 gets disintegrated into two small fragments of mass number 80 and 120. If binding energy per nucleon for parent atom is 6.5MeV and for daughter nuclei is 7 MeV and 8 MeV respectively, then the energy released (in MeV) in the decay will be:

4. A tuning fork having a frequency of 340 Hz is vibrated just above a cylindrical tube. The height of the tube is 120 cm. Water is slowly poured in it. What is the minimum height (in cm) of water required for resonance? Velocity of sound in air = 340 m/s.
5. A steel wire of length 2.0 m is stretched through 2.0 mm. The cross-sectional area of the wire is 4.0 mm². Calculate the elastic potential energy (in J) stored in the wire in the stretched condition. Young's modulus of steel = 2.0×10^{11} N m⁻².
6. The efficiency of a carnot engine is 0.6. It rejects total 20 J of heat. The work done by the engine (in J) is :
7. At a point on earth surface horizontal component of earth's magnetic field is 40 μ T and dip angle is 30°. Find the total magnetic field of earth (in μ T) at this point in nearest integer.
8. In a single slit diffraction experiment first minima for $\lambda_1 = 660$ nm coincides with the first maxima for wavelength λ_2 . Calculate λ_2 (in nm) (approximately).
9. A telescope of aperture diameter $D = 32$ mm, and objective focal length $f = 24$ cm, forms images of two distant stars in the focal plane of the objective lens. If the stars have a minimum resolvable angular separation according to the Rayleigh criterion, then what is the distance, x (in μ m), between the centers of the images in the focal plane? (Assume light of wavelength 550 nm is used.)
10. A metal sphere cools at rate of 3° C/min when its temperature is 50°. Find the rate of cooling of sphere at 40°C (temperature of surrounding is 25°C) in °C/min.

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. The distance of spherical nodes from nucleus for the given orbital are

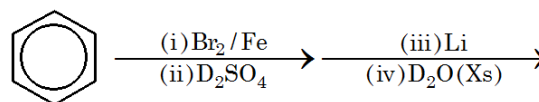
$$\psi_{\text{radial}} = \frac{1}{9\sqrt{2}} \left(\frac{Z}{a_0} \right)^{3/2} [(\sigma^2 - 4\sigma + 3)] \exp(-\sigma/2)$$

where a_0 & Z are the constants and $\sigma = \frac{2Zr}{a_0}$

- (A) Zero, infinity (B) $\frac{a_0}{Z}, \frac{1}{2} \frac{a_0}{Z}$
 (C) $\frac{3}{2} \frac{a_0}{Z}, \frac{1}{2} \frac{a_0}{Z}$ (D) $\frac{a_0}{Z}, \frac{3}{2} \frac{a_0}{Z}$
2. A definite amount of gaseous hydrocarbon was burnt with just sufficient amount of O_2 . The volume of all reactants was 600 ml, after the explosion the volume of the products [$CO_2(g)$ and $H_2O(g)$] was found to be 700 ml under the similar conditions. The molecular formula of the compound is

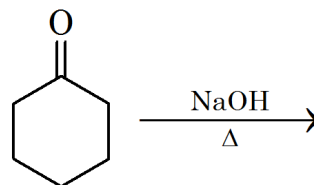
- (A) C_3H_8 (B) C_3H_6
 (C) C_3H_4 (D) C_4H_{10}

3. Major product of given reaction is :



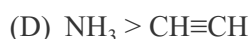
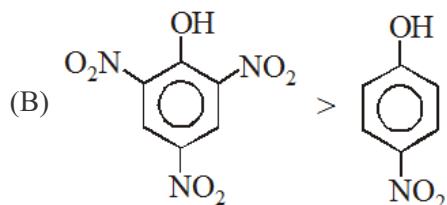
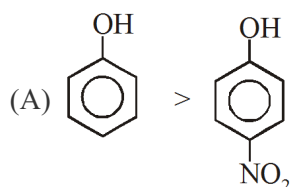
- (A) (B)
 (C) (D)

- 4.



- (A) (B)
 (C) (D)

5. Identify correct acidic strength order ?



6. Which of the following has maximum boiling point ?



7. Hardness of water is due to dissolved impurities of—

(A) calcium and magnesium salts

(B) barium and magnesium salts

(C) calcium and strontium salts

(D) sodium and potassium salts

8. The most soluble salt in water is



9. On strong heating AgNO_3 we get



10. Solid AgNO_3 is slowly added to a solution that has 0.0001 M each in NaCl , NaBr and NaI . K_{sp} of AgCl is 1.7×10^{-10} , K_{sp} of AgBr = 3.3×10^{-13} , K_{sp} of AgI = 1.5×10^{-16} . The concentration of Ag^+ required to initiate the precipitation of AgCl is



11. In a cubic close packing, the unit cell has...

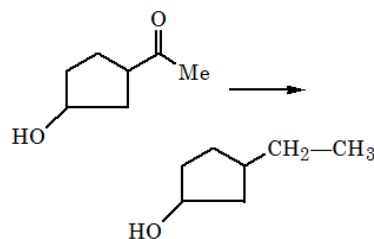
(A) 4 tetrahedral voids each of which is shared by four unit cells.

(B) 4 tetrahedral voids within the unit cell

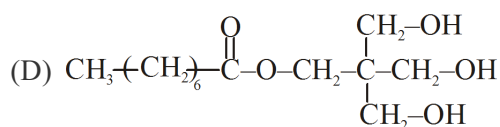
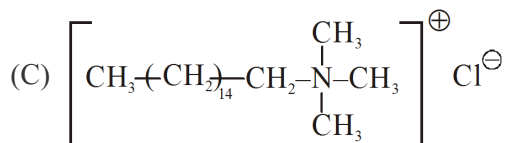
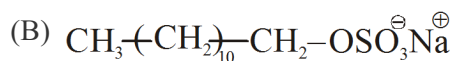
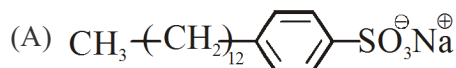
(C) 8 tetrahedral voids each of the which is shared by four unit cells

(D) 8 tetrahedral voids within the unit cells

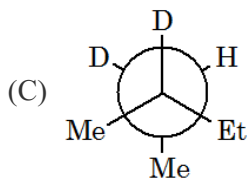
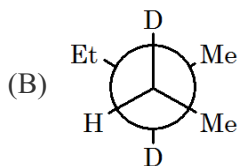
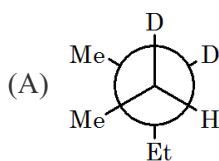
12. The appropriate reagent for the transformation is



13. Which of the following is an example of cationic detergent ?



14. Identify major product of reaction of (E)-3-methyl-2-pentene with D_2/Ni .



(D) Both (A) and (C)

15. Which of the following is responsible for transition of heredity character ?

(A) Glucose (B) Fructose
(C) DNA (D) Haemoglobin

16. Which carbohydrate is used for silvering of mirror ?

(A) Sucrose (B) Cellulose
(C) Starch (D) Glucose

17. Which of the following statements is not correct ?

(A) Nitrogen forms triple bonds whereas phosphorus does not exist as $P \equiv P$.
(B) The N-N bond is stronger than P-P bond
(C) Red phosphorus is less reactive than white phosphorus
(D) Sulphur exhibits catenation.

18. The role of calcination in metallurgical process is
(A) To remove moisture
(B) To decompose carbonate into oxide
(C) To remove volatile organic matter (impurity)
(D) All of these

19. The electronegativity of the following elements increases in the order

(A) C, N, Si, P (B) N, Si, C, P

(C) Si, P, C, N (D) P, Si, N, C

20. Which of the following is incorrect ?

(A) Classical smog occurs in cool humid climate
(B) Photochemical smog is reducing smog
(C) Mixture of smoke, fog and SO_2 form classical smog
(D) Photochemical smog occurs in warm, dry and sunny climate

SECTION-II : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

- For a real gas obeying Vander Waal equation, the values of critical pressure and critical temperature are 73.89 atm and 300 K. What will be the volume occupied, only by the molecules of 24 moles of the gas in millilitres.
($R = 0.0821 \text{ atm-lit/mol-kelvin}$)
- The pressure over pure solid A is 60 mm of Hg at a certain temperature T and the pressure over pure solid D is 80 mm of Hg at same temperature T, if A and D dissociate as
 $A(s) \rightleftharpoons B(g) + C(g)$ and
 $D(s) \rightleftharpoons B(g) + E(g)$
then the total pressure over a mixture of excess of A(s) and D(s) at same temperature will be (in mm of Hg)
- A real gas is subjected to an adiabatic process causing a change in state from (3 bar, 50 L, 500 K) to (5 bar, 40 L, 600 K) against a constant pressure of 4 bar. The magnitude of enthalpy change for the process is :
- 
How many stereoisomers are possible for this compound.
- Number of linear molecules among the following :
 I_3^- , NO_2 , H_2O , H_2S , NO_2^+ , HN_3 , CH_3NCS
- Number of Species that will show Fac-Mer isomerism among the following.
(i) $[PtCl_2(NH_3)_2]$
(ii) $[Co(NH_3)_3(NO_2)_3]$
(iii) $[Co(en)_3]^{3+}$
(iv) $[CoCl_2(en)_2]^+$
(v) $[Pt(NH_3)_3(NO_2)_3]^+$
(vi) $[CoCl_2(tren)]Cl$
- Silane (SiH_4) burns in air forming SiO_2 and H_2O as
 $SiH_4(g) + 2O_2(g) \rightarrow SiO_2(s) + 2H_2O(g)$
what is the value of ΔG° of reaction if ΔG_f° for the formation of $SiH_4(g)$, $SiO_2(s)$ and $H_2O(g)$ are +52.3, -805, -228.6 kJmol^{-1} respectively. (in kJmol^{-1})
- Osmotic pressure of a solution containing 2 gm dissolved protein per 300 cm^3 of solution is 20 mm of Hg at 27°C . The molecular mass of protein is (in gm mol^{-1}) : ($R = 0.08 \text{ L-atm/mol-K}$)
- For consecutive first order reaction:
 $A \xrightarrow{k_1} B \xrightarrow{k_2} C$, at 300 K
 $k_1 = 2 \times 10^{-3} \text{ s}^{-1}$ and $k_2 = 5 \times 10^{-5} \text{ s}^{-1}$
(Given : $\ln 2 = 0.7$, $\ln 10 = 2.5$)
The time at which $[B]$ will be maximum is—
- Total number of electrons having $\ell = 2$ in Fe^{2+} according to Aufbau principle ($n + 1$ rule)

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

- The value of $\sum_{r=1}^9 \sin^2 \frac{r\pi}{10}$ is :
(A) 1 (B) $\frac{5}{2}$
(C) 5 (D) $\frac{9}{2}$
- If the tangent and the normal to the hyperbola $x^2 - y^2 = 4$ at a point cut off intercepts a_1, a_2 on X-axis and b_1, b_2 on Y-axis respectively then the value of $a_1 a_2 + b_1 b_2$ is :
(A) 1 (B) -1
(C) 0 (D) 4
- If the point $(\cos \theta, \sin \theta)$ lies in the right angle between the rays $y = |x - 1|$, then θ belongs to
(A) $(0, \frac{\pi}{2})$ (B) $(0, \frac{\pi}{4})$
(C) $(0, \frac{\pi}{6})$ (D) $(0, \frac{\pi}{3})$
- If the system of linear equations $x + y + z = 6$, $x + 2y + 3z = 14$ and $2x + 5y + \lambda z = \mu$ ($\lambda, \mu \in \mathbb{R}$) has a unique solution, then :
(A) $\lambda \neq 8$ (B) $\lambda = 8, \mu \neq 36$
(C) $\lambda = 8, \mu = 36$ (D) $\lambda \neq 8, \mu \neq 36$

- The sum of the coefficients of even power of x in the expansion of $(1 + x + x^2 + x^3)^5$ is :
(A) 256 (B) 128
(C) 512 (D) 64

- If $A = \begin{bmatrix} \sin \alpha & -\cos \alpha & 0 \\ \cos \alpha & \sin \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then A^{-1} is

equal to :

- (A) $2A^T$ (B) A
(C) $\text{adj } A$ (D) None of these
- Let $f(x) = x^3 + bx^2 + ax + 12$ ($b \neq 0$). If $(-3, 2)$ is the largest possible interval for which $f(x)$ is decreasing function then the value of 'a' is :
(A) 3 (B) 9
(C) -9 (D) -18
- $\lim_{x \rightarrow 0} \left(\tan \left(\frac{\pi}{4} - x \right) \right)^{\frac{1}{x}}$ is equal to :
(A) 1 (B) e
(C) e^2 (D) e^{-2}
- The sum of values of x satisfying the equation $(31 + 8\sqrt{15})^{x^2-3} + 1 = (32 + 8\sqrt{15})^{x^2-3}$ is :
(A) 3 (B) 0
(C) 2 (D) 4

10. $\int \frac{x^9 dx}{(4x^2 + 1)^6}$ is equal to :

(A) $\frac{1}{5x} \left(4 + \frac{1}{x^2}\right)^{-5} + C$

(B) $\frac{1}{5} \left(4 + \frac{1}{x^2}\right)^{-5} + C$

(C) $\frac{1}{10} (1 + 4x^2)^{-5} + C$

(D) $\frac{1}{10} \left(4 + \frac{1}{x^2}\right)^{-5} + C$

11. If $f(x) = x^3 \operatorname{sgn}(x)$, then :

(A) f is derivable at $x = 0$

(B) f is continuous but not derivable at $x = 0$

(C) L.H.D. at $x = 0$ is 1

(D) R.H.D. at $x = 0$ is 1

12. $\int_{-3}^3 x^8 \{x^{11}\} dx$ is equal to (where $\{.\}$ is fractional part function)

(A) 3^8 (B) 3^7 (C) 3^9 (D) 3^{10}

13. The solution of the differential equation $(x + 2y^3) \frac{dy}{dx} = y$ is :

(A) $\frac{x}{y^2} = y + c$ (B) $\frac{x}{y} = y^2 + c$

(C) $\frac{x^2}{y} = y^2 + c$ (D) $\frac{y}{x} = x^2 + c$

14. If a variable X takes value 0, 1, 2, ..., n with frequency ${}^nC_0, {}^nC_1, \dots, {}^nC_n$ respectively, then the $\operatorname{Var}(X)$ is :

(A) $\frac{n^2 - 1}{12}$ (B) $\frac{n}{2}$

(C) $\frac{n}{4}$ (D) None of these

15. $(p \vee q) \wedge \sim p$ is equivalent to :

(A) $\sim p \wedge q$ (B) $p \wedge q$

(C) $p \wedge (\sim q)$ (D) $\sim p \wedge \sim q$

16. Let N denotes the set of all natural numbers and R be the relation on $N \times N$ defined by :

$(a, b) R (c, d) \Rightarrow ad(b+c) = bc(a+d)$, then R is :

(A) Reflexive, Symmetric but not Transitive

(B) Symmetric, Transitive but not Reflexive

(C) Transitive, Reflexive but not Symmetric

(D) Equivalence

17. Negation of the statement $p \rightarrow (q \wedge r)$ is :

(A) $\sim p \rightarrow \sim (q \wedge r)$ (B) $\sim p \vee (q \wedge r)$

(C) $(q \wedge r) \rightarrow p$ (D) $p \wedge (\sim q \vee \sim r)$

18. If P is any arbitrary point on the circumcircle of the equilateral triangle of side length ℓ units, then $|\overrightarrow{PA}|^2 + |\overrightarrow{PB}|^2 + |\overrightarrow{PC}|^2$ is always equal to :

(A) $2\ell^2$ (B) $2\sqrt{3}\ell^2$

(C) ℓ^2 (D) $3\ell^2$

19. The equation of a plane which passes through the point of intersection of lines $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ and $\frac{x-1}{3} = \frac{y-2}{1} = \frac{z-3}{2}$ and at greatest distance from origin is :

(A) $4x + 3y + 5z = 25$ (B) $4x + 3y + 5z = 50$

(C) $3x + 4y + 5z = 49$ (D) $x + 7y - 5z = 2$

20. Let $f(x) = \cos 2x \cdot \cos 4x \cdot \cos 6x \cdot \cos 8x$,
then $\lim_{x \rightarrow 0} \frac{1 - (f(x))^3}{5x^2}$ is equal to :
(A) 36 (B) 56 (C) 60 (D) 66

SECTION-II : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

1. Sum of roots of the equation $\sin^{-1}x - \cos^{-1}x = \sin^{-1}(3x - 2)$ is
2. Locus of mid point of the chords of contact of $x^2 + y^2 = 2$ from the points on the line $3x + 4y = 10$ is a circle with centre P. If O is the origin then OP is equal to :

3. The number of common chords of the parabolas $x = y^2 - 6y - 17$ and $y = x^2 - 6x + 1$ are :
4. If $x + \frac{1}{x} = \frac{1 + \sqrt{5}}{2}$, then the value of $x^{90} + \frac{1}{x^{90}}$ is :
5. The value of the sum $\sum_{k=1}^{\infty} \sum_{n=1}^{\infty} \frac{k}{2^{n+k}}$ is :
6. If x_1, x_2, x_3 and x_4 are the roots of the equation $x^4 + x^3 + 2 = 0$, then the value of $\prod_{i=1}^4 (3x_i - 1)$ is equal to :
7. A coin is tossed 7 times. Then the probability that atleast 4 consecutive heads appear is :
8. Let $f(x) = x^2 - 2x$, $x \in \mathbb{R}$ and $g(x) = f(f(x) - 1) + f(5 - f(x))$, then the minimum integral value in the range of $g(x)$ is :
9. The area of region enclosed by curves $y = x^2$ and $y = \sqrt{|x|}$ is :
10. Let $f(x) = (x - 1)(x - 2)(x - 3) \dots (x - n)$ and $f'(n) = 5040$ then the value of n is :

Space for Rough Work